Book III — Cartridge and Expansion Reference

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BMSX

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Conventions

Addresses are byte addresses unless stated otherwise. Register bit numbers refer to the named word; structure offsets are measured in bytes. Hexadecimal quantities use uppercase digits and the suffix H. Program fragments retain the notation of their language, including BLua hexadecimal literals beginning with `0x`.

Publication revisions identify editions of this manual, not console hardware revisions.

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Reader’s guide

A BMSX cartridge may contain ROM, writable memory, mapped devices or a combination of these resources. This volume specifies how software selects a socket and accesses the card’s address windows and mailbox interface.

The cartridge chapter introduces socket selection before describing the address map, bus access, registers and interrupt service. Use Book I for the console memory map and DMA controller, and Book II for program-image records and startup.

1 Cartridge sockets and mapped card devices

The BMSX has two cartridge sockets, numbered 0 and 1. A program can execute from the ROM in one socket while using data or writable memory on the card in the other. A card may contain ROM, RAM, a word mailbox, or a combination of these devices; a RAM-only card can therefore extend a program on a second card.

Both sockets use the same cartridge addresses. To read a card, software must choose both **an address within the card** and **the socket to which the access is directed**. The address determines whether the access reaches ROM, RAM or a mapped device. The socket selection determines which card supplies that region. Changing the data selection does not change the socket supplying the program's instructions.

1.1 Card composition

ROM holds the card's read-only bytes, including any program and resource data. RAM provides writable memory at separate cartridge addresses. A mailbox adds a word port, an interrupt latch and DMA-request controls; it is described as a separate device later in this chapter. None of these devices requires the other two to be present. Consult the card specification for its installed devices and capacities.

Resource	Use	Capacity
Socket	Connection for one card	Two sockets, numbered 0 and 1
Card ROM	Program and read-only data	Card-specific, within the 512-MiB ROM window
Card RAM	Writable data	Card-specific, within the 15-MiB RAM window
Mailbox	Word exchange, interrupt and DMA-request control	DATA and CONTROL words with a local interrupt latch

The controller's STATUS register reports which sockets are occupied, not the devices or capacities fitted to a card. An occupied socket need not contain a bootable ROM. The address windows accommodate cards of different capacities; locations beyond installed ROM or RAM read as zero rather than repeating the smaller memory.

1.2 Selection and execution

Consider a program executing from socket 0, with RAM fitted to both cards. The word at `30000010H` is at byte offset 16 within either card's RAM. If software selects socket 0 and reads that address, it obtains socket 0's word. Selecting socket 1 and reading the same address obtains socket 1's word; the address itself has not changed.

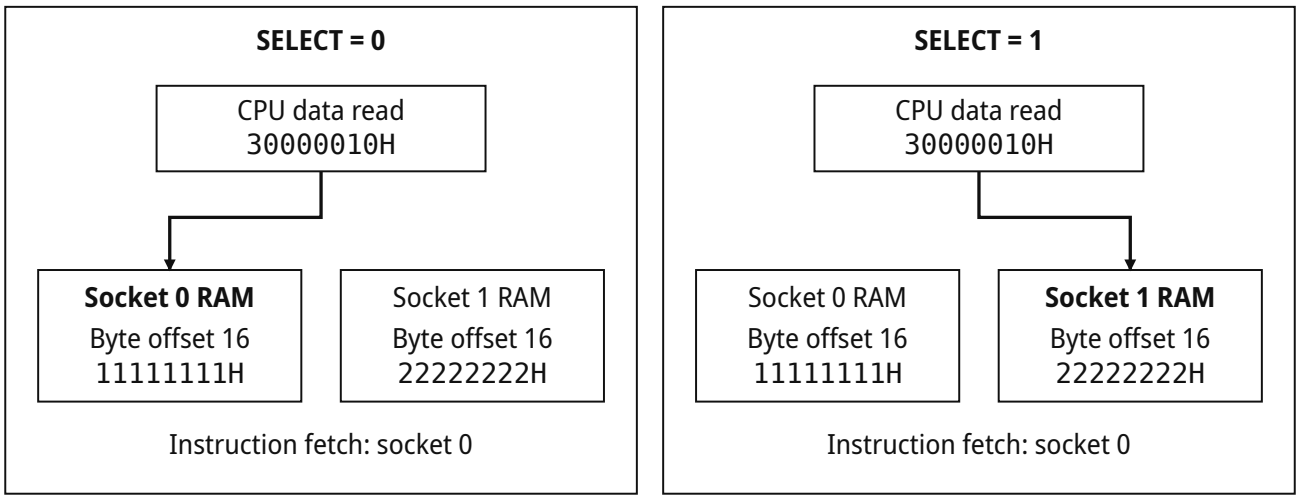


Figure 1.1: The same CPU data address reads different card RAM when SELECT changes; instruction fetch remains on socket 0. The RAM contents shown are example values.

The ordinary data selector is the controller's SELECT register. Its value applies to subsequent ordinary cartridge reads and writes until software changes it. Selection neither copies memory between cards nor resets either card. In the example, the two words keep their separate values as SELECT changes.

Instruction fetch uses a separate **execution-socket latch**. A program entered from socket 0 continues to fetch its cartridge instructions from socket 0 even when SELECT is changed to 1 for data access. This is what permits a running program to use the second card: selecting its data does not redirect the next instruction to the same address on that card.

A cartridge-targeted `CP0.EXEC` samples SELECT and establishes the execution socket for the new entry. Since cartridge addresses are identical in both sockets, insertion in the other socket does not by itself require relocation. A cartridge-function closure contains a function-record address, not a socket selection; calling it uses the current execution socket. See Book II, §3.3.3 and Book II, §1.13.

DMA also distinguishes the address from the socket. Its source and destination request fields can select different sockets, allowing a direct copy from one card to the other. Those endpoint selections are independent of the CPU data selector; their encodings and request conditions are given under DMA between sockets.

1.3 Cartridge controller

The controller has two installed-I/O registers: SELECT chooses the socket for ordinary data accesses, and STATUS reports socket occupancy and that selection. They are separate from the registers on either card, so both remain accessible regardless of which socket is selected.

1.3.1 Controller registers

Address	Name	Access	Controller-reset value
`08010420H`	`SELECT`	Read/write	`00000000H`
`08010424H`	`STATUS`	Read-only	Installed-cartridge presence bits; bit 16 clear

`SELECT` retains the complete 32-bit word written to it. Bit 0 selects the socket; bits 31:1 are retained and read back without affecting selection.

Register	Bits	Meaning
`SELECT`	0	Ordinary socket selection: 0 selects socket 0; 1 selects socket 1.

Register	Bits	Meaning
SELECT	31:1	Retained raw bits; no effect on socket selection.
STATUS	0	1 when socket 0 contains a cartridge.
STATUS	1	1 when socket 1 contains a cartridge.
STATUS	16	Ordinary selected socket: 0 for socket 0; 1 for socket 1.
STATUS	15:2, 31:17	Read as zero.

1.3.2 Accessing a card

For ordinary data access:

1. Read STATUS to determine which sockets are occupied. Use the card's own specification to determine its memory and devices.
2. Write 0 or 1 to SELECT for the intended card.
3. Access the required address within that card's ROM, RAM or MMIO window.

For the two-RAM example in Figure 1.1, these are the complete selection and read operations. Both cards are assumed to contain at least 20 bytes of RAM, with the illustrated words already stored at offset 16.

```
local card_word<const>: *word = 0x30000010
mem[0x08010420] = 0
local first = *card_word        -- 0x11111111 from socket 0.
mem[0x08010420] = 1
local second = *card_word       -- 0x22222222 from socket 1.
```

SELECT remains 1 after the second read. Any later ordinary cartridge access therefore uses socket 1 until SELECT is changed again. Code still executing from socket 0 must select socket 0 again before an ordinary data read of its own card.

1.3.3 Selection and presence

A SELECT write takes effect for subsequent ordinary cartridge accesses. Both cards retain their device state, interrupt latches and DMA-request levels across the selection change.

Instruction fetch and the explicit DMA socket selections bypass SELECT without changing it. STATUS bit 16 continues to report ordinary data selection, not the socket used by an instruction fetch or by a DMA endpoint.

The presence bits are independent of selection. Selecting an empty socket changes STATUS bit 16 while leaving bits 1:0 unchanged. Reads from that socket return zero and writes have no effect.

1.3.4 Controller reset

Controller reset clears the entire SELECT word and resets each present card. The DMA-request levels then reflect the reset state of those cards; empty sockets assert no requests. Occupancy is unchanged, so STATUS may still report present cards with socket 0 selected. Controller reset does not itself raise or acknowledge a central interrupt. The storage and mailbox reset effects are specified below.

1.4 Card storage and MMIO access

Each selected card uses the following address windows. ROM and RAM offsets are byte offsets from their respective bases; the card's installed capacity may be smaller than the window.

Region	First address	Exclusive limit	Window size
ROM	10000000H	30000000H	512 MiB
RAM	30000000H	30F00000H	15 MiB

Region	First address	Exclusive limit	Window size
MMIO	`30F00000H`	`31000000H`	1 MiB

Absent or unpopulated ROM and RAM locations read as zero. Smaller memories are not mirrored through the remaining window. ROM writes are ignored; a RAM write takes effect only when its complete width fits the installed memory.

ROM and RAM scalar values are little-endian: the lowest-addressed byte occupies bits 7:0. The starting address of a scalar read selects the region for the complete value. If fewer installed bytes remain than the requested width, available bytes occupy the low-order lanes and missing high-order lanes read as zero. A sequence of individual byte reads instead selects the region at each byte's address, so such a sequence can cross into the next window.

Operation	Width, capacity, tail, or reset behavior
ROM/RAM byte read	Returns the addressed installed byte, otherwise zero.
ROM/RAM 16-bit read	At a one-byte tail, returns that byte in bits 0–7 and zero in bits 8–15. Returns zero beyond capacity.
ROM/RAM 32-bit read	Assembles remaining bytes into successive low-order byte lanes; unavailable high-order lanes are zero. Returns zero beyond capacity.
RAM write	An 8-, 16-, or 32-bit write occurs only when its starting address selects RAM and its entire width fits installed RAM. A partial-width tail write changes nothing.
MMIO 8-/16-bit write	Ignored, including on a card with a mapped device.
MMIO 32-bit write	Writes the complete word at the exact MMIO-relative byte offset. Has no effect when no mapped device is present.
Logical card reset	Resets the mapped device, if present. Retains RAM contents and leaves ROM unchanged.

MMIO reads return zero when no device is present. Otherwise, the device decodes the byte offset from `30F00000H` without prior rounding to a word boundary. An 8-bit read selects the decoded word's byte lane using address bits 1:0. A 16-bit read selects its low or high half using address bit 1. A 32-bit read returns the complete word. Use the device's register map for the word selected by each offset; the mailbox's read decoding is given below. These read-lane rules do not enable byte or halfword stores.

Card RAM initially contains zero. A logical card reset retains its contents.

1.5 Boot selection

At startup, the BIOS examines socket 0, then socket 1, selecting each present card and reading its ROM header. It enters the first card whose ROM passes the startup checks. A RAM-only card, or a card whose ROM does not meet those requirements, is skipped. This entry establishes the execution socket as described under Selection and execution.

Book II, §3.4 specifies the ROM header layout.

1.6 Cartridge mailbox

A mailbox adds a retained DATA word and separate controls for notification and DMA requests. Software can use DATA for a value defined by the card's own protocol. Writing it replaces the previous word; DATA is a single word, not a queue. CONTROL can request an interrupt from that socket or set the request levels that permit DMA transfers involving the card.

Data and notification are separate operations. Writing DATA does not request an interrupt, and an interrupt request does not change DATA. A protocol that uses both can therefore write its value first, then request notification. The card's programming specification defines the value and what its recipient should do with it. A DMA completion reports the end of transport, not an application response from the card.

The mailbox's **local pending latch** records whether its interrupt has been requested. Clearing this latch rearms the next interrupt edge. A separate pending bit in the installed interrupt controller records that edge for CPU delivery; servicing an

interrupt requires both acknowledgements, as explained below. Mailbox transfers and interrupts are optional operations, not prerequisites for ordinary ROM or RAM access.

The mailbox base is `30F00000H` on the selected card. It has two retained 32-bit words, DATA and CONTROL; STATUS reads the local pending latch and IRQ_ACK provides its acknowledgement action. All offsets below are relative to the mailbox base.

1.6.1 Mailbox programming sequence

For a protocol that publishes one DATA word and requests an interrupt:

1. Select the card containing the mailbox. The following accesses use that card's MMIO window.
2. Starting with the local pending latch clear, write the word to DATA.
3. Write CONTROL with IRQ_TRIGGER set, retaining any DREQ or other CONTROL fields that must remain unchanged. This sets local pending and raises the socket's central interrupt source.
4. In the service routine, handle DATA according to the card protocol, clear local pending through IRQ_ACK to rearm the edge, and acknowledge the socket's central pending bit separately. Select the correct card before accessing its mailbox.

For DMA, program the required DREQ levels instead; both endpoint requests and the channel's address, count and control registers must be ready before START. The mailbox fields are defined next, and DMA between sockets shows how they connect to a transfer.

1.6.2 Register map

Address	Name	Access	Bits / word-write action	Reset read
`30F00000H`	DATA	R/W	Retains all 32 bits; write replaces the complete word	`00000000H`
`30F00004H`	CONTROL	R/W + trigger	Bits 31:1 retained; bit 0 requests an IRQ and is excluded from readback	`00000000H`
`30F00008H`	STATUS	R; writes ignored	Bit 0 IRQ_PENDING; bits 31:1 zero	`00000000H`
`30F0000CH`	IRQ_ACK	R=0; W action	Any nonzero word clears local pending; zero has no effect	`00000000H`

Read decoding ignores the low two offset bits. Offsets `+00H` through `+03H` therefore select DATA, `+04H` through `+07H` select CONTROL, and `+08H` through `+0BH` select STATUS. The read width then selects the byte lanes as described under Card storage and MMIO access. Every other offset group reads zero, including `+0CH` through `+0FH`.

Use aligned word stores for the mailbox actions. Only `+00H`, `+04H`, and `+0CH` perform an action. Writes to STATUS and other aligned offsets are ignored; byte and halfword writes do not perform mailbox actions.

1.6.3 DATA and CONTROL

DATA holds the last complete word written to it. Its read does not consume the word. A DATA write leaves CONTROL and the local pending latch unchanged.

CONTROL has the following defined fields:

Bits	Mask	Name	Meaning on a CONTROL write
0	`00000001H`	IRQ_TRIGGER	Requests a local interrupt-pending transition. Not retained by the write.

Bits	Mask	Name	Meaning on a CONTROL write
1	`00000002H`	DREQ_READ	Sets the DREQ_READ level.
2	`00000004H`	DREQ_WRITE	Sets the DREQ_WRITE level.
31:3	`FFFFFFF8H`	Unnamed	Retained exactly as written; no defined IRQ or DREQ contribution.

A CONTROL write replaces bits 31:1, including unnamed bits. Preserve any of those fields that must remain unchanged when constructing a new value. Bit 0 performs the IRQ_TRIGGER action and reads as zero; it is not stored in CONTROL. Setting this bit can request notification without a later trigger-clear write.

1.6.4 Interrupt pending and acknowledgement

An IRQ_TRIGGER write sets local IRQ_PENDING. Only its clear-to-set transition generates a new interrupt edge. Once pending is set, further trigger writes leave it set without generating additional edges.

To rearm the edge, write any nonzero word to IRQ_ACK. A subsequent IRQ_TRIGGER can then generate another edge. A zero IRQ_ACK write and a CONTROL write with IRQ_TRIGGER clear both leave local pending unchanged. IRQ_ACK changes neither DATA nor CONTROL. It acknowledges the mailbox latch, not the central interrupt-controller latch.

1.6.5 DREQ levels and reset

DREQ_READ and DREQ_WRITE follow retained CONTROL bits 1 and 2 independently. Software may assert either level, both, or neither. The levels remain asserted until replaced by a CONTROL write or cleared by reset. IRQ_TRIGGER and IRQ_ACK do not replace this programming, and unnamed CONTROL bits do not affect DREQ.

Reset clears DATA, CONTROL and local pending. All four register ports then read zero and both DREQ levels are deasserted.

1.6.6 Interrupts: two latches, two acknowledgements

Socket 0 raises central interrupt bit 9 (`00000200H`); socket 1 raises bit 10 (`00000400H`). These are distinct from each card's local STATUS bit 0. The central FLAGS port is `08000000H`, ACK is `08000004H` and MASK is `08000008H`. A central ACK clears the pending bits selected by its written ones. It does not clear the card's local pending latch.

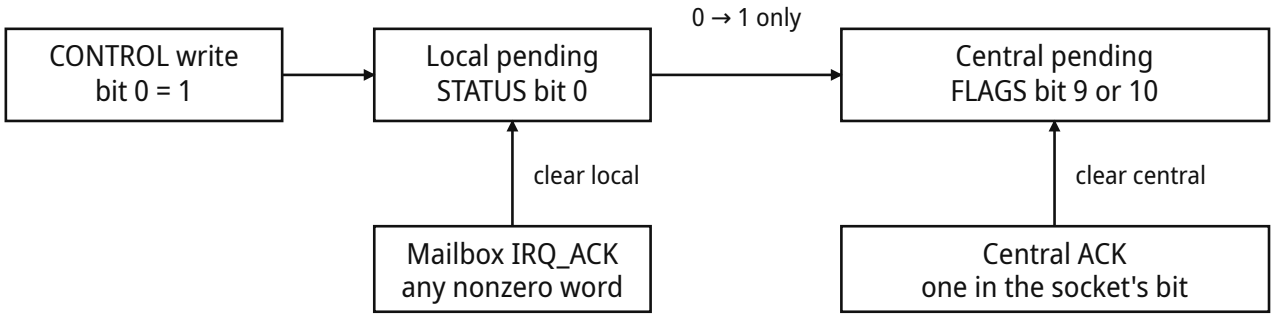


Figure 1.2: A card-local edge and its separate central pending bit

Central pending records an edge even with its MASK bit clear. MASK affects delivery, not pending storage. A FLAGS read changes neither latch. Conversely, mailbox IRQ_ACK clears only the local latch; an already pending central bit remains until central ACK. Neither latch counts repeated events.

For one socket, starting with both pending latches clear:

Accepted operation	Local pending	Central socket bit
CONTROL = `1`	1	1
Central ACK = socket mask	1	0
CONTROL = `1` again	1	0
Local IRQ_ACK = `1`	0	0
CONTROL = `1` again	1	1
Local IRQ_ACK = `1`	0	1
Central ACK = socket mask	0	0

The third row demonstrates why acknowledging only the central controller does not rearm a mailbox edge. When handling both sockets, select the intended card before touching its local registers and acknowledge only the central bits owned by that handler. The complete CPU-enable, `irq(flags)` and predicate-wait route is in Book I, §3.2.

1.7 DMA between sockets

To copy ROM data from socket 0 into RAM on socket 1, the DMA channel needs two addresses and two socket selections. Its read address chooses the source location in card ROM, while its write address chooses the destination in card RAM. The source and destination request fields supply their socket selections independently. The CPU's SELECT register need not change during the copy.

Each request field has two effects: it selects a request condition that must be satisfied before a block starts, and, for a cartridge request, it selects that endpoint's socket. `READ_REQUEST` controls the source; `WRITE_REQUEST` controls the destination. In either field, selectors 7–10 mean:

Selector	Request condition	Endpoint socket override
7	Socket 0 `DREQ_WRITE`	Socket 0
8	Socket 0 `DREQ_READ`	Socket 0
9	Socket 1 `DREQ_WRITE`	Socket 1
10	Socket 1 `DREQ_READ`	Socket 1

These selections are independent of one another and of SELECT. An endpoint with any of these four selectors may address ROM, card RAM or mailbox MMIO; the address selects the region within the chosen socket. In particular, the READ or WRITE name in a request condition does not itself restrict the endpoint's address range.

1.7.1 Request levels and block operation

Both endpoint requests must be asserted for a block to start. Mailbox CONTROL bit 1 drives DREQ_READ and bit 2 drives DREQ_WRITE. Since the requests are levels, one CONTROL write can keep a condition asserted for successive blocks; there is no separate request pulse for each word. They do not report FIFO occupancy. An absent card, or one without a mailbox, asserts neither cartridge request. `FORCE` (selector 0) is always asserted and `DISABLED` (selector 15) is never asserted.

At the start of a block, DMA captures its read and write addresses, remaining count, control and block length. The transfer cost elapses before the endpoint accesses occur. At completion, DMA reads each word and then writes it, then updates the channel's addresses and count. A remaining count of zero completes the channel. A request that drops after a block has started holds off the next block; it does not shorten the block already in progress.

Selectors other than 7–10 supply no independent cartridge socket selection. A cartridge endpoint then uses SELECT **at the time of its access**, not the selection present at START. Changing SELECT during an active block can therefore redirect that endpoint. Keep selection unchanged until completion or use an explicit cartridge request selector.

1.7.2 Four-word control value

For the socket-0 ROM to socket-1 RAM copy, use source selector 8 and destination selector 9. Increment both addresses and choose a four-word block:

`CONTROL` field	Value	Contribution
Read increment, bit 0	1	`00000001H`
Write increment, bit 1	1	`00000002H`
Read request, bits 5:2	8	`00000020H`
Write request, bits 9:6	9	`00000240H`
Block words minus 1, bits 13:10	3	`00000C00H`
Combined `CONTROL`		**`00000E63H`**

Each increment advances four bytes per word. The resulting `00000E63H` requires socket 0 DREQ_READ and socket 1 DREQ_WRITE. Once both levels permit the block to start, the CPU's current selection cannot reroute either endpoint.

1.7.3 Transfer timing

DMA timing uses the 33,868,800 Hz system clock. A contiguous N-word cartridge span costs `8N + 4` cycles, whether it addresses card ROM, card RAM or mailbox MMIO. A main-RAM span costs `N + 1` cycles. Two cartridge endpoints have additive costs even when they select different sockets:

Transfer, wholly within each endpoint's region	Cycles	Four words	Sixteen words
Cartridge ↔ main RAM	`8N + 4`	36	132
Cartridge ↔ cartridge	`16N + 8`	72	264
Cartridge ↔ installed I/O	`8N + 4`	36	132

The four-word ROM(socket 0) → RAM(socket 1) block therefore takes `(8 × 4 + 4) + (8 × 4 + 4) = 72` cycles once started. Incrementing across a ROM, RAM or MMIO boundary starts another span and incurs setup again. Add request waiting, channel competition and CPU setup or polling to obtain the complete operation time. Book I, Chapter 4 gives the general channel programming sequence.

1.7.4 Two-card programming example

The following sequence copies four ROM words from socket 0 to RAM on socket 1. Assume a bootable card in socket 0 with ROM and a mailbox, and a data card in socket 1 with at least 32 bytes of RAM and a mailbox. The sequence owns DMA0, both mailbox controls, socket-1 RAM bytes 16–31 and central DMA0 pending bit 0. No other writer changes these resources or the selected request levels. Completion is polled with interrupt delivery masked. The transfer only needs the two programmed mailbox request levels; it does not require an interrupt request from either card.

Declare four words in socket 0's ROM, assert the read request on socket 0 and the write request on socket 1, then program the idle channel:

```
rodata seed: word[4] = {
    0x01234567, 0x89abcdef, 0x55aa33cc, 0xf00dcafe,
}
mem[0x08000008] = 0           -- Mask delivery; retain pending source bits.
mem[0x08000004] = 1           -- Acknowledge this owner's old DMA0 pending bit.
mem[0x08010420] = 0
mem[0x30f00004] = 2           -- Socket 0 DREQ_READ.
mem[0x08010420] = 1
mem[0x30f00004] = 4           -- Socket 1 DREQ_WRITE.
mem[0x0800000c] = seed        -- DMA0 READ_ADDR: socket-0 ROM.
mem[0x08000010] = 0x30000010  -- DMA0 WRITE_ADDR: socket-1 RAM.
mem[0x08000014] = 4           -- Four words, not four bytes.
mem[0x08000018] = 0x00000e63  -- Increment both; requests 8/9; four-word block.
mem[0x08000020] = 1           -- START.
while (mem[0x0800001c] & 1) ~= 0 do
end
```

The explicit endpoint overrides keep the source on socket 0 and destination on socket 1 while ordinary CPU selection remains 1. After normal completion, DMA STATUS is 2 and COUNT is zero. READ_ADDR is `seed + 16`, WRITE_ADDR is `30000020H`, and the masked DMA0 pending bit is set. Before consuming the data, check for new bus faults using the procedure in Book I, §2.10.1; DMA DONE does not imply valid endpoint accesses.

Read socket 1's destination while it remains selected:

```
local destination<const>: *word = 0x30000010
local first = destination[0]
local last = destination[3]
mem[0x30f00004] = 0           -- Deassert socket-1 requests after completion.
mem[0x08010420] = 0
mem[0x30f00004] = 0           -- Deassert socket-0 requests.
mem[0x08000004] = 1
```

The destination words are `01234567H`, `89ABCDEFH`, `55AA33CCH` and `F00DCAFEH`; `first` and `last` read the first and fourth. Socket-0 RAM is unchanged. Card RAM, including the destination, retains its contents across a logical card reset. The block takes 72 system cycles, plus CPU setup, request waiting, polling and result reads.

1.8 Access faults

CPU alignment checks precede the cartridge bus access. Ordinary halfword accesses require two-byte alignment and word accesses four-byte alignment. A misaligned ordinary load or store enters its address-error exception before reaching the bus. A new mapped fault during an ordinary load enters a data-bus-error exception before committing the read value; a new fault during an ordinary store enters the same exception. A bus-level zero return is not a successful ordinary CPU load of zero. See Book I, §2.7.

Installed I/O and cartridge MMIO have different mapped-access rules. The controller's SELECT and STATUS registers occupy installed-I/O word slots. For mapped accesses to these two registers:

Access	Result
Aligned 32-bit read	Reads the complete register word.
8-bit read at the aligned register address	Reads the low byte.
Unaligned 32-bit access or 8-bit read at another byte lane	`UNALIGNED_IO` (2).
16-bit read, 8-bit store or 16-bit store	`UNALIGNED_IO` (2).
Aligned 32-bit store to SELECT	Replaces the selection word.
Aligned 32-bit store to STATUS	`READ_ONLY` (3); STATUS is unchanged.

Faulting mapped reads return zero and faulting stores are suppressed. In contrast to controller STATUS, a word store to mailbox STATUS is ignored without a READ_ONLY fault.

At cartridge addresses, the mapped access uses the card's offset decoding and byte lanes rather than the installed-I/O alignment rule. Mailbox byte and halfword stores are ignored. Its word stores decode the exact relative offset; an unaligned mailbox store therefore changes no register. An ordinary CPU halfword or word instruction is still subject to the CPU alignment check described above.

The entire access must fit inside the cartridge aperture. Its highest admitted starting addresses are `30FFFFFCH` for a word, `30FFFFFEH` for a halfword and `30FFFFFFH` for a byte. An access that crosses the final boundary raises `UNMAPPED` (1), returning zero or suppressing the store. Within the aperture, an absent card returns zero and ignores writes without a fault; ROM writes are also ignored.

DMA uses these mapped-access rules, but continues its programmed reads, writes and count progression after a fault. Check for new bus faults as well as DONE before using the result. The first outstanding bus fault retains its code, address and access; later faults do not replace it. See Book I, §2.8 for the fault registers and acknowledgement procedure.